Chapter 6 — Chemical Energetics

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. ΔH equals: (a) H reactants - H products (b) Mass products (c) H products - H reactants (d) Temperature
2. Exothermic reaction has: (a) Positive ΔH (b) Negative ΔH (c) Zero ΔH always (d) No heat
3. Activation energy is shown by: (a) Profile peak (b) Final level (c) Initial level (d) Zero line
4. Standard temperature: (a) 300 K (b) 273 K (c) 298 K (d) 100 K
5. Calorimeter heat formula: (a) q = m/cΔT (b) q = mΔT (c) q = cΔT/m (d) q = mcΔT
6. Hess's law says ΔH is: (a) Always zero (b) Path dependent (c) Path independent (d) Only for gases
7. Bond breaking: (a) Releases energy (b) Destroys atoms (c) Absorbs energy (d) No energy
8. Lattice energy forms solid from: (a) Gaseous ions (b) Atoms only (c) Molecules (d) Aqueous ions

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is ΔH negative for combustion?
2. How does a catalyst affect activation energy and ΔH?
3. Why is Hess's law useful?
4. How is q = mcΔT used to find molar enthalpy change?
5. Why does hydration release energy?
6. How does the Born-Haber cycle use Hess's law?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain enthalpy change, energy profile diagrams and standard enthalpy changes. **(4)**
(b) Explain Hess's law and its use in calculating enthalpy changes. **(4)**
Q.4 (a) Describe calorimetry and bond energy calculations. **(4)**
(b) Explain hydration energy, lattice energy and the Born-Haber cycle. **(4)**

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